🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 2 Standard

🎯 TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.
Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Convert Measurements within the Same System | Conversion factor | Unit ratio | Convert | Measurement system | Double number line | Ratio table

👆 *Tap any word to see what it means and a picture.*

1. Feet into inches stays inside the ______ measurement system.

2. The ratio that trades one unit for another is a ______ ______.

3. A ratio comparing an amount to exactly ______ unit is a unit ratio.

4. To change a measurement to another unit without changing the amount is to ______.

5 Inches and feet belong to the customary ______ ______; centimeters and meters to the metric one.

6 Two lines whose matching ticks show the same amount in two units form a ______ ______ ______.

7 A table whose every column holds the same comparison is a ______ ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many cup-scoops of broth does the recipe need, and how do you know?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Convert Measurements within the Same System** — Rewriting a measurement in a different unit of the SAME system, such as feet into inches or liters into milliliters.
  Convertir medidas dentro del mismo sistema — Reescribir una medida en otra unidad del MISMO sistema, como pies a pulgadas o litros a mililitros.

- ☐ **Conversion factor** — The ratio that links two units in the same measurement system, such as 12 inches to 1 foot. Within one system it is exact.
  Factor de conversión — La razón que relaciona dos unidades del mismo sistema de medición, como 12 pulgadas a 1 pie. Dentro de un mismo sistema es exacta.

- ☐ **Unit ratio** — A ratio that compares an amount to exactly 1 unit of another quantity, like 12 inches to 1 foot.
  Razón unitaria — Una razón que compara una cantidad con exactamente 1 unidad de otra cantidad, como 12 pulgadas por 1 pie.

- ☐ **Convert** — To change a measurement from one unit to another without changing how much it is.
  Convertir — Cambiar una medida de una unidad a otra sin cambiar la cantidad que representa.

- ☐ **Measurement system** — A family of units that go together — customary units like inches and feet, or metric units like milliliters and liters.
  Sistema de medidas — Una familia de unidades que van juntas: unidades usuales como pulgadas y pies, o métricas como mililitros y litros.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The unit ratio is ____ cups to 1 quart. Cups are ____ than quarts, so I ____ to convert. 3 × 4 = ____ cups. The amount of broth did not change — only the ____ did.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.

Evidence: Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Reasoning: This evidence connects the definition of convert measurements within the same system to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The unit ratio is ____ cups to 1 quart. Cups are ____ than quarts, so I ____ to convert. 3 × 4 = ____ cups. The amount of broth did not change — only the ____ did.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements within the Same System
2. **Notes 2:** Conversion factor
3. **Notes 3:** Unit ratio
4. **Notes 4:** Convert
5. **Notes 5:** Measurement system
6. **Notes 6:** Double number line
7. **Notes 7:** Ratio table
8. **Watch for this mistake:** *A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size — multiplying every time. Converting 36 inches to feet by multiplying gives 36 × 12 = 432 feet, taller than a 40-story building. Ask first which unit is smaller: it takes MORE small units to make the same amount, so going to a smaller unit multiplies, and going to a larger unit divides.*
9. **Try It 1:** *— Every unit ratio says how many of the smaller unit fit inside exactly 1 of the larger one.*
10. **Try It 2:** *— Conversions in this lesson stay inside one system — customary with customary, metric with metric.*